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Cross-sectional and longitudinal associations between subcutaneous adipose tissue thickness and dual-energy X-ray absorptiometry fat mass

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Abstract

The present study examined cross-sectional and longitudinal relationships between total and segmental subcutaneous tissue thicknesses from ultrasonography (US) and total and segmental fat mass (FM) estimates from dual-energy X-ray absorptiometry (DXA). Traditional US FM estimates were also examined. Twenty resistance-trained males (mean ± SD; age: 22.0 ± 2.6 years; body mass: 74.8 ± 11.5 kg; DXA fat: 17.5 ± 4.5%) completed a 6-week supervised resistance training programme while consuming a hypercaloric diet. Pre- and post-intervention body composition was assessed by DXA and B-mode US. Data were analysed using Pearson's correlation (*r*), Lin's correlation coefficient (CCC), paired *t*-tests, Wilcoxon signed-rank tests and Bland–Altman analysis, as appropriate. Cross-sectionally, correlations were observed between total DXA FM and total subcutaneous tissue thickness ($r = 0.88$). Longitudinally, a correlation was observed between total DXA FM changes and total subcutaneous tissue changes ($r = 0.49$, CCC = 0.38). Correlations of similar magnitudes were observed for the upper body and trunk estimates, but DXA FM changes were unrelated to subcutaneous tissue changes for the lower body and arms. Cross-sectionally, US 2-compartment FM and DXA FM were correlated ($r = 0.91$, CCC = 0.83). Longitudinally, a weaker correlation was observed ($r = 0.47$, CCC = 0.33). In summary, longitudinal associations between US and DXA are weaker than cross-sectional relationships; additionally, correlations between US subcutaneous tissue and whole-body DXA FM appear to be driven by the trunk region rather than appendages. Reporting raw skinfold thicknesses rather than FM estimates alone may improve the utility of techniques based on subcutaneous tissue thickness, such as US and skinfolds.

KEYWORDS

body composition, DEXA, DXA, skinfold calipers, skinfold thickness, ultrasonography, ultrasound

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1 | INTRODUCTION

Body composition is a key measure of health at the individual and population level (Wells & Fewtrell, 2006). Currently, the most common metric of body composition utilized for disease risk assessment is body mass index (BMI), defined as body mass (kg) divided by height squared (m^2; Johnson et al., 2017). Although BMI is a practical metric due to its quick and easy application, it solely describes the relationship between an individual's height and weight and is unable to distinguish between fat-free mass (FFM) and fat mass (FM), thus limiting its utility in active populations with relatively greater amounts of FFM compared with the general population. It has been demonstrated that directly assessing FFM and FM, instead of relying on BMI alone, is key to properly assessing metabolic and disease risk (Johnson et al., 2017; Leahy et al., 2013). As such, having a higher percentage of FFM relative to FM generally reduces one's risk for developing cardiovascular disease, diabetes, chronic kidney disease and many forms of cancer (Afshin et al., 2017).

Body composition variables are also highly relevant to athletic performance, as success in sport can often be influenced by an athlete's body composition (Abe et al., 2020; Claessens et al., 1999; O'Neill et al., 2016). As such, body composition estimation is frequently used to assess athletes who seek to reduce body fat and increase skeletal muscle mass. Body composition information can also be useful for the identification of potentially advantageous body phenotypes in sport and provides a means to monitor the effectiveness of a nutritional and training regimen over time (Barbieri et al., 2017).

A variety of body composition estimation techniques are employed in athletic populations, ranging from simple skinfold calipers to advanced medical devices such as dual-energy X-ray absorptiometry (DXA) scanners. Originally designed to estimate bone mineral density, DXA functions by quantifying the tissue attenuation of X-ray radiation, allowing for the identification of bone mass, FFM and FM (Shepherd et al., 2017; Toombs et al., 2012; Wells & Fewtrell, 2006). This technology is now widely used to assess body composition in research settings (Wells & Fewtrell, 2006) and is known to produce reliable body composition estimates, with accuracy compared with a criterion multi-compartment model potentially varying based on the population being assessed (Graybeal et al., 2020; Johnson et al., 2017; Toombs et al., 2012). Within the general population, it has been reported that most comparisons between DXA and criterion four-compartment models indicate a mean underestimation of body fat percentage (BF%) by DXA (Toombs et al., 2012). However, a recent comprehensive review indicated that DXA is a valid method in multi-ethnic, African-American/Black and Caucasian populations, with insufficient information currently available for Hispanic and Asian populations (Blue et al., 2021). Disparate results have been observed when determining whether leanness influences DXA's validity. Some work has reported greater underestimations of DXA BF%, as compared with a four-compartment model, with increasing leanness (Ploeg et al., 2003), while other research has indicated the opposite, with DXA overestimating BF% in the leanest athletes but underestimating BF% in individuals with relatively higher body fat (Graybeal et al., 2020). However, differences in DXA hardware and software in investigations conducted over the past three decades make broad conclusions regarding the technology's differential validity difficult to establish.

Despite its popularity, the broad utility of DXA is limited due to high equipment cost and regulatory difficulties associated with its utilization of ionizing radiation (Shepherd, Ng et al., 2017). DXA scanners are also not portable, limiting their field use. Due to these drawbacks, many health professionals, practitioners and coaches have implemented alternative assessment techniques. While not a detraction *per se*, it is important to note that standard body composition evaluation via DXA is a molecular-level assessment and does not directly estimate tissue components like skeletal muscle or adipose tissue, a distinction extensively reviewed by Heymsfield and Wang (Heymsfield et al., 1997). Briefly, a molecular-level body composition assessment categorizes all mass in the body based on broad molecular categories, such as fat, fat-free, water, mineral or protein. Therefore, molecular FM refers to all fat molecules in the body, regardless of where they occur.

Ultrasonography (US) is a medical technique that has been utilized for decades but has only recently gained popularity as a body composition assessment technique (Smith-Ryan et al., 2018). This technology functions by transmitting high-frequency sound waves that bounce back to the device to create an image when coming into contact with body tissues (Wagner, 2013). These images allow for the assessment of body composition due to the varying frequencies that different tissues reflect. There are two main types of US, amplitude mode (A-mode) and brightness mode (B-mode). B-mode US produces a 2-D image that is ideal for the assessment of body composition since it allows the user to better determine varying tissues within the image. Importantly, US devices are portable and relatively inexpensive compared with other research-grade body composition devices (Johnson et al., 2017), making them a more practical option for many organizations. However, no standardized method for using US to assess percent body fat has yet been developed, warranting further research (Smith-Ryan et al., 2018). Furthermore, it is important to note that US can be used either as a molecular-level assessment or as a tissue-level assessment. The molecular level is employed when the sum of skinfolds estimated by US is utilized in a body density equation [e.g. Jackson and Pollock's equation (Jackson & Pollock, 1978)], with the subsequent body density value inserted into a two- or three-component model equation to ultimately yield estimates of FM and FFM (Tinsley, Moore, et al., 2020; Tinsley, Rodriguez, et al., 2020). In contrast, US-derived raw skinfold thicknesses can be directly interpreted as an estimate of subcutaneous adipose tissue thickness, which is more closely related to regional tissue-level assessments.

A unique characteristic of both DXA and US is their ability to provide segmental body composition estimates (Kutáč et al., 2019; Smith-Ryan et al., 2018), thus reducing the impact of whole-body variability in tissue distribution between specific individuals or ethnic groups (Blue et al., 2021; Wells & Fewtrell, 2006).

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Additionally, segmental body composition evaluation may allow for more direct outcome monitoring when regional adaptations are desired, such as when abdominal fat is specifically targeted by a nutritional and training intervention to promote health or comply with organizational anthropometric standards. Direct comparisons between DXA and US have been previously made, with generally discordant results (Baranauskas et al., 2017; Johnson et al., 2014, 2017; Loenneke et al., 2014; Pineau et al., 2009). Baranauskas et al. (2017) found the overall agreement between US and DXA to be poor when assessing college-aged males and females using a three- or seven-site Jackson-Pollock method. Similarly, Johnson et al. (2014) also reported that US-derived estimates did not agree with DXA in the same population using the seven-site Jackson-Pollock method. Finally, Loenneke et al. (2014) determined that US was not a valid estimate of BF% when compared with DXA in female collegiate gymnasts. In contrast to these studies, Pineau et al. (2009) reported that US had good agreement with DXA in athletes of various sports. Importantly, all of these investigations examined the relationships between DXA and US at the molecular level of body composition assessment, meaning that US skinfold estimates were used in a series of equations to produce two-component (2C) model estimates of FM, FFM, or BF%. To our knowledge, no investigations have examined the relationship between raw skinfold estimates, which are reflective of subcutaneous adipose tissue, and body composition estimates produced by DXA. Due to the segmental information obtained by these methods, both whole-body and regional comparisons can be made. Additionally, it is important to note that most body composition methodology studies are conducted cross-sectionally. While these investigations are often more time-effective than longitudinal examinations, they do not provide direct evidence of the comparability of methods for tracking body composition changes over time. Thus, the purpose of the present study was to examine the cross-sectional and longitudinal relationships between total and segmental subcutaneous adipose tissue thicknesses obtained by US and total and segmental FM estimates produced by DXA.

2 | METHODS

2.1 | Overview

Twenty healthy resistance-trained males (mean ± SD; age: 22.0 ± 2.6 years; height: 179.1 ± 7.0 cm; body mass (BM): 74.8 ± 11.5 kg; DXA body fat: 17.5 ± 4.5%) completed a 6-week intervention that included 3 days/week of supervised resistance training and consumption of a hypercaloric diet, as previously described (Smith et al., 2021). The study protocol was approved by the Texas Tech University Institutional Review Board, and all participants provided informed consent prior to participation. The current report is a secondary analysis reporting cross-sectional and longitudinal relationships between DXA and US body composition variables.

2.2 | Body composition assessment

Before and after the 6-week intervention, DXA scans were performed on a Lunar Prodigy scanner (General Electric) with enCORE software (v. 16.2). The scanner was calibrated daily per manufacturer instructions. Participants were positioned using custom-made foam blocks to ensure consistent and standardized body positioning (Moore et al., 2019; Nana et al., 2012). A foam block was placed between the hand and hip bilaterally, with the hands maintaining a neutral position. An additional block and strap enabled consistent positioning of the feet. A trained researcher manually adjusted region of interest lines within the enCORE software to designate body segments (i.e. arms, trunk and legs; Tinsley, Moore, et al., 2020; Tinsley, Rodriguez, et al., 2020). In our laboratory, these procedures have produced intraclass correlation coefficient (ICC) values from 0.974 to 0.996 and root-mean-square coefficient of variation (RMS-CV) from 2.6 to 5.4% for total and segmental FM estimates.

B-mode US was used to estimate subcutaneous tissue thickness at several body sites. While participants stood upright, transmission gel was applied to the seven measurement locations specified by Jackson and Pollock (1978). A linear-array probe (Lumify L12-4, Philips Healthcare) connected to an electronic Android tablet was used. Minimal pressure was applied by the transducer to avoid tissue compression, and a single image was taken at each measurement location. The gain was maintained at the manufacturer's set value (50) for all images. Depth was adjusted as needed to visualize tissue at each site but remained consistent for all measurements at a particular site within a given participant. A single trained technician performed all assessments in this study and another technician with expertise in US assessment manually reviewed each image for quality assurance. The technician performing the US assessments previously demonstrated an ICC of 0.997 and a RMS-CV of 4.0% for subcutaneous tissue estimates obtained on separate days. Raw single-layer tissue thicknesses – reflective of the layer of skin and subcutaneous adipose tissue – were provided by the US software interface (MuscleSound®). These single-layer thicknesses were utilized in the analyses reporting the relationships between DXA FM variables and US thickness estimates. However, in the analysis incorporating US 2C FM values, the single-layer thicknesses were doubled by the analysis software (MuscleSound®) to promote consistency with the values obtained by skinfold calipers. This was necessary due to the usage of the seven-site sum of skinfolds (SS) and body density (D_b) equations of Jackson and Pollock (1978):

$$\mathrm{SS} = \Sigma\,(\text{chest} + \text{axilla} + \text{triceps} + \text{subscapula} + \text{abdomen} + \text{suprilium} + \text{front thigh})$$

$$D_b = 1.112 - (0.00043499 * \mathrm{SS}) + (0.00000055 * \mathrm{SS}^2) - (0.00028826 * \text{age}))$$

The US FM estimates were based on the Siri 2-compartment model equation (Siri, 1961):

$$\mathrm{FM} = \mathrm{BM} * \left(\frac{4.95}{D_b} - 4.5\right).$$

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2.3 | Statistical analysis

Data were analysed using R (v. 4.0.2; R Core Team, 2019). Relationships between DXA and US variables were examined using Pearson product-moment correlations (*r*) and when the same variable or metric (i.e. % change) was being compared between methods, Lin's concordance correlation coefficient (CCC) was computed (Lin, 1989). Normality of data was examined via Shapiro–Wilk tests and quantile-quantile plots. When data were not normally distributed, they were transformed prior to *r* estimation using the *bestNormalize* package (Peterson & Cavanaugh, 2019). Correlation coefficients and associated 95% confidence intervals are presented for *r* and CCC. Raw data are presented in the results to aid interpretation.

Changes across time in DXA and US variables were evaluated via paired-samples *t*-tests or Wilcoxon signed-rank tests when changes were not normally distributed. For comparisons of DXA and US-derived FM estimates, additional validity analysis was performed due to the same variable (i.e. FM) being examined for both methods. These analyses included Bland–Altman analysis (Bland & Altman, 1986) and estimation of the total error (TE). The primary R packages utilized for analysis were *DescTools* (Signorell, 2019) and *ggplot2* (Wickham, 2016).

3 | RESULTS

3.1 | Changes in outcome variables

Increases in DXA total mass (4.0 ± 2.1 kg; $p < 0.001$), total FM (1.6 ± 1.6 kg; $p < 0.001$), upper body FM (1.2 ± 1.1 kg; $p < 0.001$), leg FM (0.4 ± 0.5 kg; $p < 0.001$), arm FM (0.2 ± 0.2 kg; $p < 0.001$) and trunk FM (1.0 ± 1.1 kg; $p < 0.001$) were observed. Additionally, increases in US-derived total skinfold thickness (7.3 ± 9.9 mm; $p = 0.004$), upper body skinfold thickness (8.1 ± 10.6 mm; $p = 0.003$) and trunk skinfold thickness (8.0 ± 10.9 mm; $p = 0.004$) were detected. However, no changes in lower body skinfold thickness (−0.7 ± 4.2 mm; $p = 0.65$) or arm skinfold thickness (0.1 ± 1.1 mm; $p = 0.66$) were identified. Individual changes in DXA FM variables and US subcutaneous tissue thicknesses are displayed in Figure 1.

3.2 | Relationships between DXA FM and US subcutaneous thickness

In the cross-sectional analysis, correlations were observed between whole-body DXA FM and total subcutaneous tissue thickness ($r = 0.88$ [95% CI: 0.72, 0.95]), upper body DXA FM and upper body subcutaneous tissue thickness ($r = 0.79$ [95% CI: 0.53, 0.91]), lower body DXA FM and lower body subcutaneous tissue thickness ($r = 0.64$ [95% CI: 0.27, 0.84]), DXA FM in the arms and arm subcutaneous tissue thickness ($r = 0.73$ [95% CI: 0.42, 0.89]) and DXA FM in the trunk and trunk subcutaneous tissue thickness ($r = 0.76$ [95% CI: 0.47, 0.90]).

Longitudinally, a significant correlation was observed between total DXA FM changes and total subcutaneous thickness changes ($r = 0.49$, CCC = 0.38; Figure 2). Correlations of similar magnitudes were observed for the upper body ($r = 0.43$, CCC = 0.38) and trunk ($r = 0.44$, CCC = 0.34), although the confidence intervals for some of these estimates crossed zero. In contrast, DXA FM changes were unrelated to changes in subcutaneous tissue thicknesses for the lower body ($r = 0.26$, CCC = 0.09) and arms ($r = 0.21$, CCC = 0.12).

FIGURE 1 Individual changes in DXA FM and ultrasound-derived subcutaneous thickness variables. The black lines represent individual participant changes, and the red line represents the mean change for each variable. Increases in all DXA variables were observed ($p < 0.001$ for each). For US, increases in total skinfold thickness (i.e. all sites; $p = 0.004$), upper body skinfold thickness ($p = 0.003$) and trunk skinfold thickness ($p = 0.004$) were observed; however, no changes in lower body skinfold thickness ($p = 0.65$) or arm skinfold thickness ($p = 0.66$) were identified

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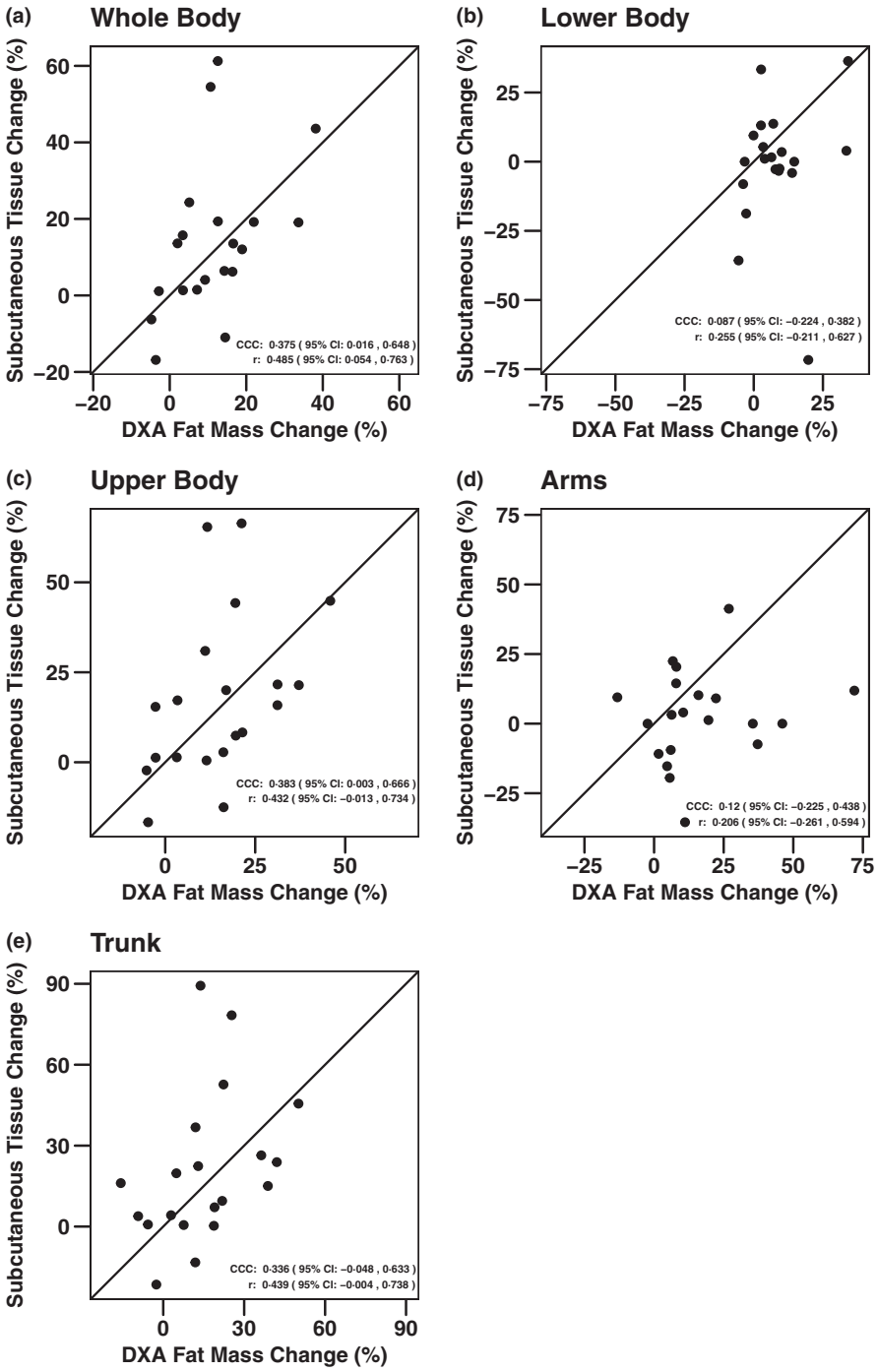

FIGURE 2 Relationship between longitudinal changes in DXA FM and ultrasound-derived subcutaneous thickness. Percent changes for single-layer tissue thicknesses and DXA FM variables are plotted for the whole body (a) and different body regions, including the lower body (b), upper body (c), arms (d) and trunk (e). CCC, concordance correlation coefficient; CI, confidence interval; DXA, dual-energy X-ray absorptiometry; *r*, Pearson's product-moment correlation; US, ultrasound

3.3 | Relationships between DXA and US whole-body FM

In the cross-sectional analysis of baseline data, 2C FM estimates obtained by DXA and US were correlated (r = 0.91, CCC = 0.83; Figure 3). The difference between FM estimates was 2.2 ± 2.1 kg, with DXA producing a higher value in all participants except one. The TE between DXA and US FM estimates was 3.0 kg. Bland–Altman analysis indicated no proportional bias in the cross-sectional comparison of FM estimates (Figure 4).

Longitudinally, changes in DXA and US FM were also correlated, although the relationship was weaker than that observed cross-sectionally (r = 0.47, CCC = 0.33; Figure 3). Bland–Altman analysis indicated proportional bias such that FM changes tended to be overestimated by US relative to DXA when FM increases were larger (Figure 4). The TE between DXA and US FM changes was 1.8 kg.

4 | DISCUSSION

The purpose of the present study was to examine the cross-sectional and longitudinal relationships between total and segmental subcutaneous adipose tissue thickness obtained by US and total and segmental

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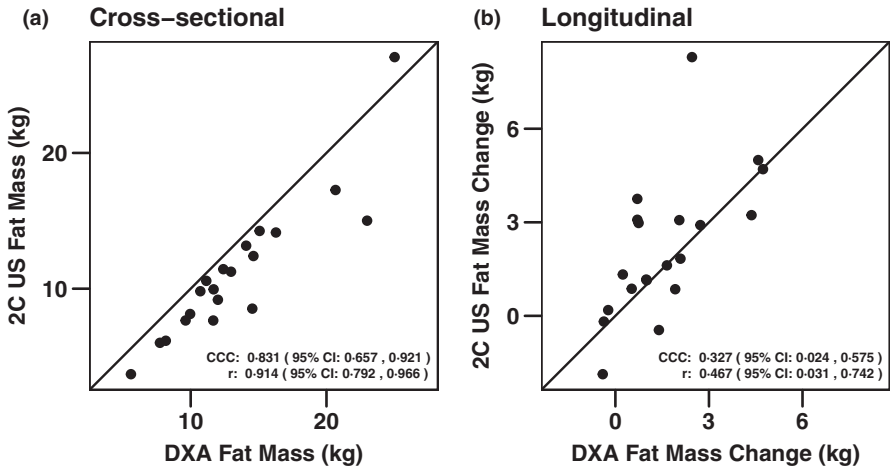

FIGURE 3 Cross-sectional and longitudinal associations between DXA and US FM estimates. Cross-sectional comparisons of baseline whole-body FM values (a) and comparisons of changes in whole-body FM between baseline and the end of the study period (b) are displayed. CCC, concordance correlation coefficient; CI, confidence interval; DXA, dual-energy X-ray absorptiometry; *r*, Pearson's product-moment correlation; US, ultrasound

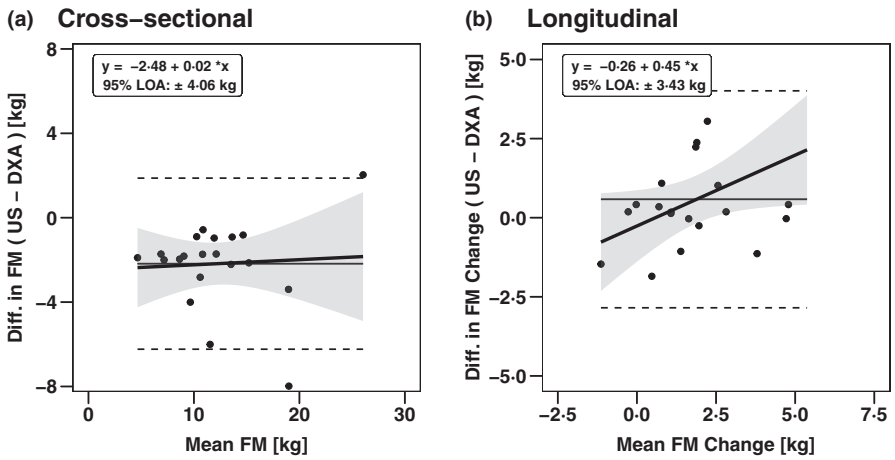

FIGURE 4 Bland–Altman analysis of cross-sectional and longitudinal DXA and US FM estimates. Bland–Altman analysis is displayed, with the solid diagonal line representing the relationship between the difference in body composition estimates- calculated as the ultrasound-derived estimate minus the DXA estimate- and the average of the ultrasound and DXA estimates. The shaded regions around the diagonal line indicate the 95% confidence limits for the linear regression line, the horizontal dashed lines indicate the upper and lower limits of agreement, and the horizontal solid line indicates the constant error between methods. The linear regression equation is also displayed. Cross-sectional comparisons (a) and comparisons of changes between baseline and the end of the study period (b) are presented. DXA, dual-energy X-ray absorptiometry; FM, fat mass; LOA, limits of agreement; US, ultrasound

FM estimates produced by DXA. The major findings of this analysis were: (1) Cross-sectional associations between US subcutaneous tissue thicknesses and DXA FM estimates were stronger than longitudinal associations (r = 0.64 – 0.88 vs. 0.21 – 0.49); (2) Longitudinally, the significant correlation between total US subcutaneous tissue and whole-body DXA FM appeared to be driven by correlations in the trunk region rather than the appendages; (3) When comparing molecular-level FM estimates from DXA and US, significant cross-sectional and longitudinal correlations were observed; however, DXA produced higher FM values in all participants except one, and proportional bias between methods was observed for FM changes.

The stronger cross-sectional, rather than longitudinal, relationship between DXA and US is noteworthy since previous studies have only examined the relationship between methods cross-sectionally (Baranauskas et al., 2017; Johnson et al., 2014, 2017; Pineau et al., 2009). These investigations have reported the association between whole-body DXA-derived FM or BF% and US-derived estimates, typically using Pearson correlations (r). For this comparison in the present study, an r value of 0.91 was observed cross-sectionally; however, the r value was only 0.47 was observed when comparing FM changes longitudinally. This is unsurprising, as the range of raw values cross-sectionally was larger than the range of change values longitudinally. If previous investigations had included longitudinal monitoring alongside cross-sectional comparison, we speculate that similar findings of poorer longitudinal than cross-sectional validity would have been observed.

Nonetheless, the existing cross-sectional data still provides preliminary information regarding the relationship between DXA and US body composition estimates as they are typically implemented. Pineau et al. (2009) compared DXA and US-derived body fat percentages in 93 highly trained athletes and found that US estimates of body fat percentage were highly correlated with those of DXA in both females (r = 0.97, SEE = 1.79) and males (r = 0.98, SEE = 0.96). Additional studies, however, reported a general lack of agreement

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between US and DXA (Baranauskas et al., 2017; Johnson et al., 2014, 2017), although interpretation of what constitutes an acceptable level of agreement has also differed between research groups. For example, Baranauskas et al. (2017) compared DXA and multiple skinfold equations using US-based tissue thicknesses for estimating BF% in 76 college-aged males and females. The researchers demonstrated significant group-level differences between DXA and all skinfold equations, with DXA producing higher BF% values (3.5–5.2%, on average). However, strong correlations (r = 0.78 to 0.82) between the assessments were also identified. Likewise, Johnson et al. (2014, 2017) conducted two similar studies comparing DXA and US body composition estimates. In the first study, a moderately significant correlation between US and DXA BF% was observed in a sample of 84 college-aged males and females (r = 0.66). When split by gender, a strong correlation for men (r = 0.84) and a moderate correlation for women (r = 0.65) were identified. However, DXA produced BF% estimates that were approximately 4% higher on average compared with US. The subsequent investigation, conducted in a sample of 74 college-aged males and females, also reported significantly higher FM estimated by DXA compared with US. The cross-sectional correlation for FM (r = 0.91) in the present study fell near the higher end of the range of correlations for FM and BF% (r = 0.66–0.98) reported by Johnson et al. (2014, 2017). As mentioned, the lower longitudinal correlation for FM in the present study (r = 0.47) may indicate that, if longitudinal examinations had been included in prior studies, correlations between DXA and US may have been lower than was reported cross-sectionally.

A unique aspect of the present study was the direct comparison of DXA FM with raw subcutaneous tissue thicknesses, rather than solely with 2C FM estimates produced by inserting the US-derived tissue thicknesses into a series of prediction equations. This comparison bridges two levels of body composition assessment, as DXA FM is considered a molecular-level entity (i.e. an estimation of the molecular fat content of the body) while subcutaneous adipose tissue thickness more closely resembles an organ/tissue-level assessment (Tinsley, Moore, et al., 2020; Tinsley, Rodriguez, et al., 2020; Wang et al., 1992). Previously, researchers have entered the sum of skinfolds into body density predictions, most frequently those developed by Jackson and Pollock (1978), with the body density value subsequently entered into a separate equation, such as the Siri equation (Siri, 1961) to yield a 2C FM estimate. Notably, both the Jackson-Pollock and Siri 2C equations were produced using hydrodensitometry values, meaning that the series of equations ultimately converts an organ/tissue-level entity (subcutaneous adipose tissue thickness) to a molecular-level whole-body entity. These conversions result in a less granular view of body composition since distinct values are summed and subsequently viewed as a single entity. Although this may be the most frequently used method of handling tissue thicknesses, it is possible that retaining the tissue thickness values in their original form could improve the utility of these values, particularly since they are a direct reflection of a potential outcome of interest (i.e. regional adipose tissue deposition). Retaining distinct thickness values allows regional changes to be tracked over time and may provide insight into specific areas of the body that may be targeted by diet and exercise interventions. Additionally, the traditional series of equations relies on several assumptions which could ultimately introduce error and weaken the validity – or usefulness – of the final value. Alternatively stated, although the typical utilization of tissue thicknesses seeks to equate US or skinfold calipers with a molecular-level assessment, these methods could instead be viewed as distinct, complementary techniques that yield their own meaningful output. In the present study, similar to the comparison of FM estimates, we observed a much stronger correlation between US total subcutaneous thickness and DXA total FM cross-sectionally (r = 0.88) as compared longitudinally (r = 0.49). This supports the importance of not only examining the cross-sectional agreement of body composition techniques but also comparing the detected changes longitudinally. Tracking changes longitudinally will better determine the true agreement between methods as the body adapts to various nutritional or training stimuli. Longitudinal monitoring also has clear applications for growth and development, tracking the progression of disease states and adaptations to lifestyle interventions.

While there may be advantages to tracking raw tissue thicknesses, there are also distinct limitations associated with this method. For example, only subcutaneous adipose tissue, along with the skin unless subtracted, is assessed. This may present interpretive challenges since not all adipose tissue in the body is stored subcutaneously; however, this problem could potentially be remedied by pairing subcutaneous tissue thicknesses with a metric of total central adiposity, such as a simple waist circumference, thereby providing a more complete picture. An advantage of such a method is its feasibility, portability and low cost relative to more expensive laboratory-based measures such as DXA.

There are several clinical applications of the present work. First, the purpose of body composition estimation is important to consider. If the goal of an assessment is a simple one-time evaluation of an individual's anthropometric characteristics compared with reference values, numerous methodologies may be adequate for this purpose. However, if longitudinal tracking is a goal, a consideration of the ability of a method to sufficiently detect changes over time is warranted. As demonstrated, differences between cross-sectional and longitudinal agreement can be observed when comparing specific methods. Second, when performing longitudinal assessments, it is imperative to use the same method of body composition estimation consistently over time; however, even when the same technology is used consistently, the observed body composition change values may still not be identical. While it is generally recognized that using values from different technologies at different time points introduces error and severely limits certainty of observed changes, the present work indicates that different changes may be observed between technologies even when using the same method consistently over time. Third, the actual measurements taken by a body composition assessment technology should be considered. While most common technologies attempt to utilize molecular-level assessment – yielding metrics of FM, FFM and BF% – they vary widely in the variables that are used to predict these body composition outcomes. For example, DXA relies

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on differential attenuation of X-rays through all body tissues while US considers only regional tissue thickness. Other methods, such as circumference-based manual or digital anthropometry, do not assess any internal body components. Fourth, while comparisons between techniques are often necessary, viewing distinct technologies as complementary – rather than striving to equate them – is an alternative perspective that recognizes the uniqueness and utility of the variables each technique is directly assessing.

Primary limitations of the present analysis are the relatively small, all-male sample and the short interval between longitudinal measurements. The potential for sex differences is noteworthy given that some previous comparisons of DXA and US demonstrated apparently better agreement (i.e. higher correlations or lower SEE values) in males as compared with females (Johnson et al., 2014; Pineau et al., 2009). In future, longer interventions conducted in varied populations and contexts (i.e. weight loss, maintenance, weight gain) may provide additional information regarding the relationship between DXA and US body composition metrics and better inform the use of these technologies.

In conclusion, the present study demonstrated a cross-sectional relationship between total and segmental US-derived tissue thicknesses and total and segmental DXA whole-body FM; furthermore, there was an association between changes in US total tissue thickness and DXA whole-body FM, which was driven by the trunk region rather than the appendages. Additionally, cross-sectional correlations between US-derived 2C FM and DXA FM were observed, with weaker correlations observed longitudinally. While the traditional method of using US body composition estimates involves converting this organ/tissue-level measure to a molecular-level estimate through a series of equations, there may be merits in evaluating the raw tissue thicknesses directly. Ultimately, while DXA and US both provide meaningful information regarding body composition changes, they should be viewed as distinct complementary technologies rather than equivalent methods.

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CONFLICT OF INTEREST

The authors have no conflicts of interest to declare.

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